

SOIL TYPES

— FEEL & COMPARE

Detailed Lesson Plan · Session 1 of 3

CBSE / NCERT · Environmental Studies (EVS)

Category	Subject	Session	Duration
Soil Types – Feel & Compare	Environmental Studies (EVS)	Session 1 of 3	30 Minutes

Board	Depth	Language
CBSE / NCERT (ICSE compatible)	Beginner	English (with Hindi transliterations)

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SESSION HOOK & WARM-UP (0:00 – 0:03)

Opening Story: Arjun's Dadi and the Diwali Diyas

Open with this relatable Indian story to spark curiosity about soil and instantly connect the topic to children's cultural world.

The Story	Arjun's dadi (grandmother) is making diyas for Diwali. She needs special mitti . Arjun goes to the river — but will <i>any</i> mitti work? Let's find out together!
Warm-Up Q.	Ask the class: Have you ever touched soil? What did it feel like? Was it smooth, rough, or sticky? Where did you find it?
Bridge	Not all soil is the same — soil comes in different types, each with its own look, feel, and use. Today we become Soil Scientists!

LEARNING OBJECTIVES

By the end of this session, every learner will be able to:

LO 1 — Concept	Understand that soil is a natural material found on the Earth's surface and that it comes in different types with different properties.
LO 2 — Identify	Name at least three types of soil found in India: Sandy Soil, Clayey Soil, and Loamy Soil.
LO 3 — Senses	Describe the look, feel, and texture of each soil type using words: rough, smooth, sticky, gritty, soft, crumbly.
LO 4 — Compare	Compare two or more soil types based on colour, texture, and water retention .
LO 5 — Real-World	Connect each soil type to its Indian everyday use : sandy → beaches/construction, clayey → kumbhaar/pottery, loamy → kisan/farming.
LO 6 — Inquiry	Ask simple "why" and "what if" questions about soil, showing early scientific thinking .

Bloom's Taxonomy Mapping

Bloom's Level	Learning Objective
Remember	LO 2 — Name the three soil types
Understand	LO 1 — Grasp what soil is and why it comes in types
Apply	LO 3 — Use senses to observe and describe soil

Analyse	LO 4 — Compare soil types by properties
Evaluate	LO 5 — Match each soil to its best use
Create	LO 6 — Ask inquiry questions about soil behaviour

30-MINUTE SESSION OUTLINE

Time	Segment	Activity / Content
0:00–0:03	Hook / Warm-Up	Arjun's dadi story · Ask: "Have you touched soil? How did it feel?" · Introduce: Soil Scientists mission
0:03–0:07	What is Soil?	Show: garden soil, river mud, pot soil — all are soil! · Definition: loose natural material on Earth's surface · Plants grow in it; animals live in it; humans use it
0:07–0:12	Sandy Soil (5 min)	Image: Marina Beach / Juhu Beach / Thar Desert · Properties: light beige, rough, gritty, drains fast, does NOT clump · Activity: "Imagine rubbing beach sand!" · Uses: beaches, construction, groundnut farming
0:12–0:17	Clayey Soil (5 min)	Image: kumbhaar at chakra making diya/matka; black cotton soil Maharashtra · Properties: smooth, sticky when wet, holds water, clumps firmly · Activity: "Wet palms pressed together!" · Uses: pottery, diya-making, cotton crops
0:17–0:22	Loamy Soil (5 min)	Image: lush green farm in Punjab / vegetable garden · Properties: dark brown, soft, crumbly, holds water AND drains extra · Activity: "Soft dark mitti that crumbles gently" · Uses: wheat, dal, rice, vegetables
0:22–0:27	Compare & Feel (5 min)	Side-by-side visual comparison of all 3 types · Interactive: "Which soil for the kumbhaar? The kisan? The beach?" · Comparison chart revealed
0:27–0:30	Recap + Quiz (3 min)	3-question quiz: Q1 rough & gritty? Q2 potter's soil? Q3 best for vegetables? · Chant: "Sandy feels rough, clay sticks like glue, loamy is the best — plants love it too!"

THREE TYPES OF SOIL — FEEL & COMPARE

■ SANDY SOIL (Reti Wali Mitti)

Property	Detail	Indian Example
Colour	Light beige / pale yellow	Thar Desert sand, river bank
Texture / Feel	Rough, coarse, gritty	Like beach sand at Juhu / Marina Beach
Water Retention	POOR — drains very fast	Water disappears quickly; not good for most crops
Clumping	Does NOT form a ball	Falls apart when you try to squeeze it
Best Use	Beaches, construction, some crops	Groundnuts (moongphali), watermelon, millets

■ CLAYEY SOIL (Chikni Mitti / Kumhaar Ki Mitti)

Property	Detail	Indian Example
Colour	Red, dark grey, or reddish-brown	Red Deccan soil; black cotton soil of Maharashtra
Texture / Feel	Smooth & sticky when wet; hard & cracked when dry	Like stiff atta (dough); smooth like soap
Water Retention	Very HIGH — holds water a long time	Can cause waterlogging; cotton loves it!
Clumping	Forms a firm ball or ribbon easily	Kumbhaar shapes it into matka, diya, kulhad
Best Use	Pottery, cotton, sugarcane, paddy	Diya-making (Diwali), Ganesh/Durga clay idols

■ LOAMY SOIL (Damat Mitti / Upjaau Mitti)

Property	Detail	Indian Example
Colour	Rich dark brown to warm brown	Indo-Gangetic Plains — Punjab, UP, Bihar
Texture / Feel	Soft, crumbly, slightly gritty but not rough	Like soft damp mitti in a flower pot
Water Retention	JUST RIGHT — holds enough, drains excess	Perfect balance — India's "food basket" soil
Clumping	Forms a soft clump that breaks apart easily	Easy to dig and plant seeds in
Best Use	Farming — best for almost all crops	Wheat, dal, rice, tomatoes, methi, maize

THE BIG COMPARISON

Property	■■ Sandy	■ Clayey	■ Loamy
Colour	Light / beige	Red / dark grey	Dark brown
Texture / Feel	Rough & gritty	Smooth & sticky	Soft & crumbly
Water holding	POOR (drains fast)	VERY HIGH	JUST RIGHT
Clumping?	■ No	■ Yes — firm	■ Soft clump
Indian example	Beach sand, Rajasthan desert	Kumbhaar ki mitti, Maharashtra black soil	Kisan ka khet, Punjab / UP farms
Best for?	Beaches, construction, groundnut	Pottery, cotton, diya-making	Wheat, dal, rice, vegetables

SENSORY LANGUAGE GUIDE — FOR CLASSROOM USE

Use these rich sensory descriptions in narration and classroom discussion to bring each soil to life:

■■ Sandy	"Rough like sandpaper." / "Like rubbing fine gravel between your fingers." / "Imagine you are at Juhu Beach — feel the sand between your toes!"
■ Clayey	"Smooth like soap." / "Sticky like wet atta (dough) — but stiffer." / "Like the smooth paste a kumbhaar uses to shape a matka on the chakra."
■ Loamy	"Soft and crumbly like dry mitti biscuits." / "Rich dark soil that breaks apart gently when you squeeze it." / "The kind of soil in a kitchen garden — easy to dig, nice to hold."

■ **Key Reframe for Students: Mitti is not dirty — it is MAGICAL! Always treat soil with curiosity and respect — never as "gross" or "dirty."**

INDIAN CONTEXT REFERENCES

Category	Indian References
Occupations	Kumbhaar (potter) — matka, diya, kulhad, clay idol maker Kisan (farmer) — tilling khet, sowing wheat/dal/rice Maali (gardener) — watering terrace/balcony garden
Places	Marina Beach (Chennai) / Juhu Beach (Mumbai) — Sandy · Thar Desert, Rajasthan — Sandy Indo-Gangetic Plains: Punjab, UP, Bihar — Loamy Deccan Plateau: Karnataka, Tamil Nadu — Red clayey Black cotton soil belt: Maharashtra, MP — Clayey
Crops	Wheat (gehu) — Loamy · Cotton (kapas) — Clayey · Groundnut (moongphali) — Sandy Dal/lentils — Loamy · Paddy/rice — Loamy or Clayey · Sugarcane (ganna) — Clayey/Loamy
Festivals	Diwali — mitti ke diye (clay lamps) · Ganesh Chaturthi / Durga Puja — clay idols from river clay Vasant Panchami — farmers worship soil and seeds
Daily Life	Kulhad (clay chai cup) at roadside stalls · Matka (clay water pot) for summer · Terracotta tiles on temples · Earthen floor (mitti ka farsh) in rural homes

HANDS-ON ACTIVITY: THE FEEL & COMPARE TRAY

Materials: 3 small containers labelled **Sandy / Clayey / Loamy** · **Soil samples or substitutes** (sandpit sand, modelling clay, potting mix) · **Small jug of water** · **Observation worksheet**

Step 1	Place 3 labelled containers of soil on each group table.
Step 2	LOOK: What colour is it? Light, dark, or red?
Step 3	TOUCH: How does it feel — rough, smooth, or soft?
Step 4	SQUEEZE: Does it stay together or fall apart in your palm?
Step 5	WATER TEST: Add a few drops of water. Does water stay on top, soak in slowly, or drain fast?
Step 6	Record observations in the worksheet comparison table.
Step 7	Group discussion: "Which soil would you use to make a diya? Why?"

Soil	Colour	Feel	Clumps?	Water behaviour
■ ■ Sandy	Light beige	Rough, gritty	No	Drains immediately
■ Clayey	Red / dark grey	Smooth, sticky	Yes — firm	Stays on top / absorbs slowly
■ Loamy	Dark brown	Soft, crumbly	Soft clump	Absorbs, then drains gently

■ *Safety: Ensure soils are clean. Wash hands after. If real soil unavailable, use sand / modelling clay / commercial potting mix as substitutes.*

QUICK QUIZ — CHECK YOUR UNDERSTANDING

Q	Question	Answer
1	Which soil is rough and gritty?	Sandy Soil
2	Which soil does a kumbhaar (potter) use for diyas?	Clayey Soil
3	Which soil is best for growing vegetables?	Loamy Soil

SUCCESS CRITERIA

A learner has successfully completed this session if they can:

- ■ Name at least 3 types of soil: sandy, clayey, and loamy.
- ■ Describe how each soil type feels using at least 1 sensory word.
- ■ Match each soil type to at least 1 real-world Indian use.
- ■ Identify which soil is best for farming (loamy) and explain simply.
- ■ Compare two soil types using comparison language.
- ■ Recall the session chant — showing retention of all three soil types.

Threshold	Result
4 out of 6 criteria met	Session Objective Achieved ✓
6 out of 6 criteria met	Mastery ■

VOCABULARY GLOSSARY

Term (Hindi)	Simple Definition
Soil (Mitti)	Loose natural material on Earth's surface. Plants grow in it.
Sandy Soil	Made of large rough grains. Feels gritty. Drains water fast. Found on beaches.
Clayey Soil	Tiny particles, smooth & sticky when wet. Holds water. Used by potters (kumbhaar).
Loamy Soil	Mix of sand, clay & silt. Dark, soft, crumbly. Best soil for farming.
Texture	How something feels when you touch it — rough, smooth, gritty, sticky, soft.
Gritty / Rough	Feeling of tiny hard grains, like beach sand. Sandy soil feels this way.
Sticky / Smooth	Clings together; no bumps. Wet clayey soil feels this way.

Crumbly	Breaks apart easily into small pieces. Loamy soil feels this way.
Water retention	How well soil holds water. Clay = high; Sand = low; Loam = just right.
Kumbhaar	A potter — person who makes pots, diyas, and cups from clayey soil.
Kisan	A farmer — person who grows crops in a khet (field).

RECAP & SUMMARY

What We Learned Today!

Point 1	Soil is a natural material on Earth's surface — made of tiny rock pieces, dead plants, water, and air.
Point 2	Three types: Sandy (rough, gritty, drains fast) · Clayey (smooth, sticky, holds water) · Loamy (soft, crumbly, best for farming).
Point 3	Each soil has a different feel — sandy is like beach sand, clay is like wet atta, loamy is like soft flower-pot mitti.
Point 4	Indian uses: kumbhaar uses clayey for diyas & matkas · kisan uses loamy for wheat, dal, rice · beaches have sandy soil.
Point 5	Mitti is not dirty — it is MAGICAL! Plants grow in it, animals live in it, and people use it every day.

"Sandy feels rough, clay sticks like glue, loamy is the best — plants love it too!"

NEXT SESSIONS PREVIEW

Session	Title	Key Topics
Session 2 (In intermediate)	Soil & Water: Absorption Experiment	Pour equal water on each soil · Observe which absorbs more / drains faster · Water retention concept · Monsoon & waterlogging in India
Session 3 (In intermediate)	Soil & Living Things / Soil Care	Soil as habitat — earthworms (kenchua), ants, microbes · Composting (khad) · Soil erosion · Project: "My Soil Diary"